

Chapter 2, Section 1

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1. Let A be an Abelian group, and define the *constant presheaf* associated to A on the topological space X to be the presheaf $\mathcal{U} \mapsto A$ for all $\mathcal{U} \neq \emptyset$, with restriction maps the identity. Show that the constant sheaf $\mathcal{A}$ defined in the text is the sheaf associated to this presheaf.

Proof. Let $\mathcal{C}^+$ be the sheaf associated to the constant presheaf $\mathcal{C}$ defined above. It suffices to show $\mathcal{C}^+$ and $\mathcal{A}$ are isomorphic at the level of stalks. Fix $P \in X$, then the stalk of $\mathcal{C}^+$ at P is the same as the stalk of $\mathcal{C}$ at P , and since the restriction morphisms of $\mathcal{C}$ are identity maps, we must have $\mathcal{C}_P^+ = \mathcal{C}_P = A$. The constant sheaf $\mathcal{A}$ is defined as $\mathcal{U} \mapsto \{\text{continuous maps } \mathcal{U} \rightarrow A\}$, where A has the discrete topology. Then, we can take the stalk of $\mathcal{A}$ at x to be the direct limit of $\mathcal{A}(\mathcal{U})$ where $\mathcal{U}$ is a connected open neighborhood of x , and if $\mathcal{U}$ is connected, $\mathcal{A}(\mathcal{U}) = A$ since the image of a connected set must be connected, i.e. $\mathcal{U} \rightarrow A$ is continuous if and only if it is constant. Therefore, we have $\mathcal{A}_P = A$. Hence, $\mathcal{C}^+$ and $\mathcal{A}$ are isomorphic. $\square$

2. (a) For any morphism of sheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, show that for each point P , $(\ker \varphi)_P = \ker(\varphi_P)$ and $(\operatorname{im} \varphi)_P = \operatorname{im}(\varphi_P)$.
 (b) Show that φ is injective (respectively, surjective) if and only if the induced maps on the stalks φ_P is injective (respectively, surjective) for all P .
 (c) Show that a sequence $\dots \rightarrow \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \rightarrow \dots$ of sheaves and morphisms is exact if and only if for each $P \in X$ the corresponding sequence of stalks is exact as a sequence of abelian groups.

Proof.

- (a) It is true in general that kernels commute with limits and cokernels commute with colimits in any abelian category, since kernels and cokernels can be realized as limits and colimits, respectively.
 (b) φ is injective $\iff \ker \varphi = 0 \iff (\ker \varphi)_P = 0 \iff \ker \varphi_P = 0$.
 φ is surjective $\iff \operatorname{im} \varphi = \mathcal{G} \iff \operatorname{im}(\varphi_P) = (\operatorname{im} \varphi)_P = \mathcal{G}_P$.
 (c) The sequence $\dots \rightarrow \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \rightarrow \dots$ is exact $\iff \operatorname{im} \varphi^{i-1} = \ker \varphi^i$ for all $i \iff (\operatorname{im} \varphi^{i-1})_P = (\ker \varphi^i)_P$ for all i and $P \in X \iff \operatorname{im}(\varphi_P^{i-1}) = \ker(\varphi_P^i)$ for all i and $P \in X$ by part (a) above.

$\square$

3. (a) Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on X . Show that φ is surjective if and only if the following condition holds: for every open set $\mathcal{U} \subseteq X$, and for every $s \in \mathcal{G}(\mathcal{U})$, there is a covering $\{\mathcal{U}_i\}$ of $\mathcal{U}$, and there are elements $t_i \in \mathcal{F}(\mathcal{U}_i)$, such that $\varphi(t_i) = s|_{\mathcal{U}_i}$ for all i .
 (b) Give an example of a surjective morphism of sheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, and an open set $\mathcal{U}$ such that $\varphi(\mathcal{U}) : \mathcal{F}(\mathcal{U}) \rightarrow \mathcal{G}(\mathcal{U})$ is not surjective.

Proof.

- (a) Suppose $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a surjective morphism of sheaves. Then for any $P \in \mathcal{U}$, the induced morphism of stalks $\varphi_P : \mathcal{F}_P \rightarrow \mathcal{G}_P$ is surjective. The elements of $\mathcal{G}_P$ (and similarly $\mathcal{F}_P$) are an equivalence class of ordered pairs $(\mathcal{U}, s)$ with $\mathcal{U}$ an open neighborhood of P and $s \in \mathcal{G}(\mathcal{U})$ and the equivalence relation $(\mathcal{U}, s) \sim (\mathcal{V}, t)$ if and only if there exists some open $W \subseteq \mathcal{U} \cap \mathcal{V}$ containing P

such that $s|_W = t|_W$. If φ_P is surjective, then for every $(U, s) \in \mathcal{G}_P$ there exists some $(V, f) \in \mathcal{G}_P$ such that

$$\varphi_P(V, f) = (V, \varphi f), \quad \varphi : \mathcal{F}(V) \rightarrow \mathcal{G}(V),$$

so there exists some open subset $W \subseteq U \cap V$ such that $\varphi f|_W = s|_W$. We can repeat this process for all $x \in U$ to obtain an open cover with the desired properties, that is $\{W_x\}_{x \in U}$ of U where W_x is defined as above with an associated $f_x \in \mathcal{F}(W_x)$ such that by definition $\varphi f_x = s|_{W_x}$.

Conversely, it suffices to show $\varphi_P : \mathcal{F}_P \rightarrow \mathcal{G}_P$ is surjective for all $P \in X$. Fix $P \in X$ and consider an arbitrary element $(U, s) \in \mathcal{F}_P$. We want to show there exists some $(V, f) \in \mathcal{G}_P$ such that $(V, \varphi_P(f)) \sim (U, s)$. To that end, by assumption there exists some open subset $W \subseteq U$ and $g \in \mathcal{F}(W)$ such that $\varphi_W(g) = s|_W$, hence $\varphi_P(W, g) = (U, s)$.

- (b) We provide an example of a sheaf from open sets to sets that satisfies the condition. Let $X, Y = S^1$. Let $\pi_X : X \rightarrow S^1$ be the identity map, and let $\tau : Y \rightarrow X$ be defined by $z \mapsto z^2$ where X, Y is identified with the unit circle in the complex plane, and let $\pi_Y := \tau \circ \pi_X$. Define a sheaf $\mathcal{F}$ on S^1 for a nonempty open subset U of S^1 as

$$\begin{aligned} \mathcal{F}(U) &= \{\text{sections } s : U \rightarrow X \text{ with respect to } \pi_X\} \\ &= \{\text{continuous maps } s : U \rightarrow X \text{ such that } \pi_X \circ s = \text{id}_U\}, \end{aligned}$$

and similarly define $\mathcal{G}$ as the map that maps U to the set of sections from U to Y with respect to π_Y . If $s \in \mathcal{G}(U)$ so that $\pi_Y \circ s = \text{id}_U$, then we can compose s with τ so that

$$\pi_X \circ (\tau \circ s) = (\pi_X \circ \tau) \circ s = \pi_Y \circ s = \text{id}_U,$$

which shows $\tau \circ s$ is a section from U to X , so we can define a morphism $\tau_{\#} : \mathcal{G} \rightarrow \mathcal{F}$ induced by τ as

$$\begin{aligned} \tau_{\#}(U) : \mathcal{G}(U) &\rightarrow \mathcal{F}(U) \\ s &\mapsto \tau \circ s. \end{aligned}$$

Let U be the entire space S^1 , then $\mathcal{G}(S^1) = \emptyset$ and $\mathcal{F}(S^1) = \{\text{id} : S^1 \rightarrow X\}$, so $\mathcal{G}(S^1) \rightarrow \mathcal{F}(S^1)$ cannot be surjective. However, for any proper open subset U of S^1 , the map $\tau_{\#} : \mathcal{G}(U) \rightarrow \mathcal{F}(U)$ is surjective, thus by part (a) above $\tau_{\#}$ is a surjective morphism. □

5. Show that a morphism of sheaves is an isomorphism if and only if it is both injective and surjective.

Proof. $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is an isomorphism $\iff \varphi_P : \mathcal{F}_P \rightarrow \mathcal{G}_P$ is an isomorphism for all $P \in X \iff \varphi$ is injective and surjective by Exercise 2. □

6. (a) Let $\mathcal{F}'$ be a subsheaf of a sheaf $\mathcal{F}$. Show that the natural map of $\mathcal{F}$ to the quotient sheaf $\mathcal{F}/\mathcal{F}'$ is surjective, and has kernel $\mathcal{F}'$. Thus, there is an exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{F}' \longrightarrow 0.$$

- (b) Conversely, if $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ is an exact sequence, show that $\mathcal{F}'$ is isomorphic to a subsheaf of $\mathcal{F}$, and that $\mathcal{F}''$ is isomorphic to the quotient $\mathcal{F}$ by this subsheaf.

Proof.

- (a) The map $\mathcal{F}(U) \rightarrow \mathcal{F}(U)/\mathcal{F}'(U)$ is surjective for all open $U \subseteq X$, so $\pi : \mathcal{F} \rightarrow \mathcal{F}/\mathcal{F}'$ is surjective. To show π has kernel $\mathcal{F}'$, by Exercise 2 it suffices to show $(\ker \pi)_P = \mathcal{F}'_P$ for all $P \in X$. Each map $\mathcal{F}(U) \rightarrow \mathcal{F}(U)/\mathcal{F}'(U)$ has kernel $\mathcal{F}'(U)$, hence we have $(\ker \pi)_P = \varinjlim \mathcal{F}'(U) = \mathcal{F}'_P$.
- (b) By Exercise 2, We have the following equivalent statements:

$$\begin{aligned} 0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0 &\text{ is exact.} \\ \iff 0 \rightarrow \mathcal{F}'_P \rightarrow \mathcal{F}_P \rightarrow \mathcal{F}''_P \rightarrow 0 &\text{ is exact.} \\ \iff \mathcal{F}'_P \subseteq \mathcal{F}_P \text{ and } \mathcal{F}''_P \simeq \mathcal{F}_P/\mathcal{F}'_P \simeq (\mathcal{F}/\mathcal{F}')_P & \\ \iff \mathcal{F}' \text{ is isomorphic to a subsheaf of } \mathcal{F} \text{ and } \mathcal{F}'' \cong \mathcal{F}/\mathcal{F}'. & \end{aligned}$$

□

7. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves.

- (a) Show that $\text{im } \varphi \cong \mathcal{F} / \ker \varphi$.
 (b) Show that $\text{coker } \varphi \cong \mathcal{G} / \text{im } \varphi$.

Proof.

- (a) It suffices to show $(\text{im } \varphi)_P \simeq \mathcal{F}_P / (\ker \varphi)_P$, which again follows from the fact that for all open $U \subseteq X$, $\text{im } (\varphi(U)) \simeq \mathcal{F}(U) / \ker (\varphi(U))$.
 (b) In the same vain, $\text{coker } \varphi \simeq \mathcal{G} / \text{im } \varphi$ follows from the fact that $\text{coker } (\varphi(U)) \simeq \mathcal{G}(U) / \text{im } (\varphi(U))$.

□

8. For any open subset $U \subseteq X$, show that the functor $\Gamma(U, \cdot)$ from sheaves on X to abelian groups is a left exact functor, i.e. if $0 \rightarrow \mathcal{F}' \xrightarrow{f} \mathcal{F} \xrightarrow{g} \mathcal{F}''$ is an exact sequence of sheaves, then $0 \rightarrow \Gamma(U, \mathcal{F}') \xrightarrow{f_U} \Gamma(U, \mathcal{F}) \xrightarrow{g_U} \Gamma(U, \mathcal{F}'')$ is an exact sequence of groups.

Proof. Exactness at $\Gamma(U, \mathcal{F}')$ follows from the fact that $\mathcal{F}' \rightarrow \mathcal{F}$ is an injective morphism of sheaves if and only if $\Gamma(U, \mathcal{F}') \rightarrow \Gamma(U, \mathcal{F})$ is injective for all U . To show exactness at $\Gamma(U, \mathcal{F})$, the sequence $\mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}''$ is exact if and only if $\mathcal{F}'_P \rightarrow \mathcal{F}_P \rightarrow \mathcal{F}''_P$ is an exact sequence for all $P \in X$.

If $s \in \text{im } f_U$ then $(U, s) \in \ker g_P$ for all $P \in X$, so $(U, g(s))$ equals to 0 everywhere locally, hence $s \in g_U$. Conversely, if $s \in \ker g_U$, then by Exercise 3 there exists an open cover $\{U_i\}$ and elements $t_i \in \Gamma(U_i, \mathcal{F}')$ such that $f_U(t_i) = s|_{U_i}$. For any i, j , consider $t_i|_{U_i \cap U_j}, t_j|_{U_i \cap U_j}$. Since $f_{U_i \cap U_j} : \Gamma(U_i \cap U_j, \mathcal{F}') \rightarrow \Gamma(U_i \cap U_j, \mathcal{F})$ is injective and $f_{U_i \cap U_j}(t_i|_{U_i \cap U_j}) = f_{U_i \cap U_j}(t_j|_{U_i \cap U_j}) = s|_{U_i \cap U_j}$, we have must $t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}$, so by the sheaf property there exists $t \in \Gamma(U, \mathcal{F}')$ such that $t|_{U_i} = t_i$, so $f_U(t) = s$, hence $s \in \text{im } f_U$. □

9. *Direct Sum.* Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X . Show that the presheaf $U \mapsto \mathcal{F}(U) \oplus \mathcal{G}(U)$ is a sheaf. It is called the *direct sum* of $\mathcal{F}$ and $\mathcal{G}$, and is denoted by $\mathcal{F} \oplus \mathcal{G}$. Show that it plays the role of direct sum and of direct product in the category of sheaves of abelian groups.

Proof. Since the category of sheaves on X is a full subcategory of presheaves on X , it suffices to show it satisfies the universal property of a direct sum and of direct product in the category of presheaves of abelian groups on X , then show it satisfies the sheaf property. Let $\pi_1 : \mathcal{F} \oplus \mathcal{G} \rightarrow \mathcal{F}$, $\pi_2 : \mathcal{F} \oplus \mathcal{G} \rightarrow \mathcal{G}$ be the canonical projection morphisms, and let $\mathcal{H} \in \mathfrak{Presh}_X$ with morphisms $\tau_1 : \mathcal{H} \rightarrow \mathcal{F}$ and $\tau_2 : \mathcal{H} \rightarrow \mathcal{G}$ such that the following diagram commutes:

$$\begin{array}{ccc}
 \mathcal{H} & \xrightarrow{\tau_2} & \mathcal{G} \\
 \searrow \alpha & & \uparrow \pi_2 \\
 & \mathcal{F} \oplus \mathcal{G} & \\
 \swarrow \tau_1 & \downarrow \pi_1 & \\
 & \mathcal{F} &
 \end{array}$$

we find the find a unique morphism $\alpha : \mathcal{H} \rightarrow \mathcal{F} \oplus \mathcal{G}$ such that the diagram above commutes. For any open $U \subseteq X$, define $\alpha(U) : \mathcal{H}(U) \rightarrow (\mathcal{F} \oplus \mathcal{G})(U)$ as $\alpha(h) = (\tau_1(h), \tau_2(h))$. Since $\alpha(U)$ is unique by the same universal property for abelian groups, α must be the desired unique morphism, and since the direct sum and direct product are essentially the same for finite collections, we have shown $\mathcal{F} \oplus \mathcal{G}$ is a direct sum and product in the category of presheaves.

Now we show $\mathcal{F} \oplus \mathcal{G}$ is a sheaf. Let U be an open subset of X , and let $\{V_i\}$ be a cover of U by open sets. If $(s, t) \in \Gamma(U, \mathcal{F} \oplus \mathcal{G})$, then $(s|_{U_i}, t|_{U_i}) = (s, t)|_{U_i} = 0$ by universal property nonsense, so $s = 0$ and $t = 0$. Also, if we have elements $(s_i, t_i) \in \Gamma(V_i, \mathcal{F} \oplus \mathcal{G})$ such that for all i, j , $(s_i, t_i)|_{V_i \cap V_j} = (s_j, t_j)|_{V_i \cap V_j}$, then again $(s_i|_{U_i \cap U_j}, t_i|_{U_i \cap U_j}) = (s_j|_{U_i \cap U_j}, t_j|_{U_i \cap U_j})$ if and only if $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ and $t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}$, so by the sheaf properties of $\mathcal{F}$ and $\mathcal{G}$, there exists $s \in \Gamma(U, \mathcal{F})$ and $t \in \Gamma(U, \mathcal{G})$ such that $(s, t)|_{U_i} = (s_i, t_i)$. □

- 10. Direct Limit.** Let $\{\mathcal{F}_i\}$ be a direct system of sheaves and morphisms on X . We define the *direct limit* of the system $\{\mathcal{F}_i\}$, denoted $\varinjlim \mathcal{F}_i$, to be the sheaf associated to the presheaf $U \mapsto \varinjlim \mathcal{F}_i(U)$. Show that this is a direct limit in the category of sheaves on X .

Proof. Let $\mathcal{F}$ denote the presheaf defined $U \mapsto \varinjlim \mathcal{F}_i(U)$, and let $\mathcal{F}^+$ denote the sheaf associated to $\mathcal{F}$ with associated morphism $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$. From the direct system, we have morphisms $\mu_{ij} : \mathcal{F}_i \rightarrow \mathcal{F}_j$ and $\mu_i : \mathcal{F}_i \rightarrow \mathcal{F}$ such that for all i, j , $\mu_i = \mu_j \circ \mu_{ij}$. To show $\mathcal{F}^+$ is a direct limit in the category-theoretic sense, we must provide the data of morphisms (between sheaves) $\alpha_i : \mathcal{F}_i \rightarrow \mathcal{F}^+$ such that $\alpha_i = \alpha_j \circ \mu_{ij}$ for all i, j satisfying the following universal property: if $\mathcal{G}$ is any sheaf on X with the data of morphisms $\{\phi_i : \mathcal{F}_i \rightarrow \mathcal{G}\}$ satisfying for all i, j , $\phi_i = \phi_j \circ \mu_{ij}$, then there exists a unique morphism $\phi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\phi_i = \phi \circ \alpha_i$. Observe the following commutative diagram:

$$\begin{array}{ccccc}
 & & \phi_i & & \\
 & \nearrow & & \searrow & \\
 \mathcal{F}_i & \xrightarrow{\mu_i} & \mathcal{F} & \xrightarrow{\psi} & \mathcal{G} \\
 & \searrow & \downarrow \theta & \nearrow & \\
 & & \mathcal{F}^+ & &
 \end{array}$$

$\alpha_i := \theta \circ \mu_i$

We claim $\{\alpha_i := \theta \circ \mu_i\}$ are the desired morphisms. By the universal property of the direct limit which defines $\mathcal{F}$ and the direct system of morphisms $\{\mu_i\}$, there exists a unique morphism between presheaves $\psi : \mathcal{F} \rightarrow \mathcal{G}$ such that $\phi_i = \psi \circ \mu_i$. Then, by the universal property of the sheaf $\mathcal{F}^+$ associated to the presheaf $\mathcal{F}$, there exists a unique morphism between sheaves $\phi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\psi = \phi \circ \theta$. It remains to check $\phi_i = \phi \circ \alpha_i$. Indeed, we have

$$\phi_i = \psi \circ \mu_i = (\phi \circ \theta) \circ \mu_i = \phi \circ (\theta \circ \mu_i) = \phi \circ \alpha_i,$$

and ϕ is unique by the universal properties above. $\square$

- 11.** Let $\{\mathcal{F}_i\}$ be a direct system of sheaves on a noetherian topological space X . In this case show that the presheaf $U \mapsto \varinjlim \mathcal{F}_i(U)$ is already a sheaf. In particular, $\Gamma(X, \varinjlim \mathcal{F}_i) = \varinjlim \Gamma(X, \mathcal{F}_i)$.

Proof. We directly show the presheaf $\mathcal{F}$ defined by $\mathcal{F}(U) = \varinjlim \mathcal{F}_i(U)$ is a sheaf. Let U be an open subset of X , and let $\{V_i\}$ be open cover of U . By the noetherian hypothesis, we can assume $\{V_j\}$ to be a finite set ($1 \leq j \leq n$), then we have the following exact sequence of abelian groups

$$0 \longrightarrow \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_{j=1}^n \Gamma(V_j, \mathcal{F}_i) \longrightarrow \prod_{j,k} \Gamma(V_j \cap V_k, \mathcal{F}_i),$$

and since both products $\prod_{j=1}^n \Gamma(V_j, \mathcal{F}_i)$ and $\prod_{j \neq k} \Gamma(V_j \cap V_k, \mathcal{F}_i)$ are finite, it is equivalent to the coproduct; in particular, colimits commute with colimits, so we have the exact sequence

$$0 \longrightarrow \varinjlim \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_{j=1}^n \varinjlim \Gamma(V_j, \mathcal{F}_i) \longrightarrow \prod_{j,k} \varinjlim \Gamma(V_j \cap V_k, \mathcal{F}_i)$$

so the sheaf properties immediately follow. $\square$

- 12. Inverse Limit.** Let $\{\mathcal{F}_i\}$ be an inverse system of sheaves on X . Show that the presheaf $U \mapsto \varprojlim \mathcal{F}_i(U)$ is a sheaf. It is called the *inverse limit* of the system $\{\mathcal{F}_i\}$, and is denoted by $\varprojlim \mathcal{F}_i$. Show that this it has the universal property of an inverse limit in the category of sheaves.

Proof. Let $\mathcal{F}$ be the presheaf $U \mapsto \varprojlim \mathcal{F}_i(U)$. We have morphisms $\pi_{ij} : \mathcal{F}_i \rightarrow \mathcal{F}_j$ and $\pi_i : \mathcal{F} \rightarrow \mathcal{F}_i$ such that for all i, j , $\pi_j = \pi_{ij} \circ \pi_i$. If U is an open set, if $\{V_\alpha\}$ is an open covering of U , then we have the exact sequence of abelian groups

$$0 \longrightarrow \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_\alpha \Gamma(V_\alpha, \mathcal{F}_i) \longrightarrow \prod_{\alpha, \beta} \Gamma(V_\alpha \cap V_\beta, \mathcal{F}_i)$$

and direct limits commute with direct products, so the fact that $\mathcal{F}$ is a sheaf follows from the exact sequence

$$0 \longrightarrow \varprojlim \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_\alpha \varprojlim \Gamma(V_\alpha, \mathcal{F}_i) \longrightarrow \prod_{\alpha, \beta} \varprojlim \Gamma(V_\alpha \cap V_\beta, \mathcal{F}_i).$$

To show $\mathcal{F}$ is an inverse limit in a category-theoretic sense, if $\mathcal{G}$ is some sheaf on X with a collection of morphisms $\tau_i : \mathcal{G} \rightarrow \mathcal{F}_i$ such that $\tau_j = \pi_{ij} \circ \tau_i$ for all i, j , then we want to show there exists a unique $\theta : \mathcal{G} \rightarrow \mathcal{F}$ such that $\tau_i = \pi_i \circ \theta$. Since for each open subset U in X we have a direct system $\tau_i(U) : \Gamma(U, \mathcal{G}) \rightarrow \Gamma(U, \mathcal{F}_i)$, by the universal property of the inverse limit $\Gamma(U, \mathcal{F}) = \varprojlim \Gamma(U, \mathcal{F}_i)$, there exists a unique morphism $\theta(U) : \Gamma(U, \mathcal{G}) \rightarrow \varprojlim \Gamma(U, \mathcal{F}_i)$ such that $\tau_i(U) = \pi_i(U) \circ \theta(U)$, so we can define θ as such, then it is unique by construction. $\square$

- 14. Support.** Let $\mathcal{F}$ be a sheaf on X , and let $s \in \mathcal{F}(U)$ be a section over an open set U . The *support* of s , denoted $\text{Supp } s$, is defined to be $\{P \in U \mid s_P \neq 0\}$, where s_P denotes the germ of s in the stalk $\mathcal{F}_P$. Show that $\text{Supp } s$ is a closed subset of U . We define the *support* of $\mathcal{F}$, $\text{Supp } \mathcal{F}$, to be $\{P \in X \mid \mathcal{F}_P \neq 0\}$. It need not be a closed subset.

Proof. We show $U - \text{Supp } s$ is an open set. If $Q \in U$ such that $s_Q = 0$, then by definition of the direct limit, there exists an open neighborhood V of Q in U such that $0 = (V, s|_V) \in \mathcal{F}_Q$, hence $s_R = 0$ for all $R \in V$. $\square$

- 15. Sheaf Hom.** Let $\mathcal{F}, \mathcal{G}$ be sheaves of abelian groups on X . For any open subset $U \subseteq X$, show that the set $\text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ of morphisms of the restricted sheaves has a natural structure of an abelian group. Show that the presheaf $U \mapsto \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ is a sheaf. It is called the *sheaf of local morphisms* of $\mathcal{F}$ into $\mathcal{G}$, "sheaf hom" for short, and is denoted $\mathcal{H}om(\mathcal{F}, \mathcal{G})$.

Proof. Morphisms $f, g \in \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ define for all open $V \subseteq U$ homomorphisms between abelian groups $f(V), g(V) : \mathcal{F}(V) \rightarrow \mathcal{G}(V)$, so we can define $f + g$ as $(f + g)(V) = f(V) + g(V)$. It is obviously an abelian group with identity 0 as the zero morphism defined by $0(V) \equiv 0$.

Let U be an open subset of X , and let $\{V_i\}$ be an open cover of U . We make some clarifying remarks. If $s \in \Gamma(U, \mathcal{H}om(\mathcal{F}, \mathcal{G})) = \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$, then s is a natural transformation of functors $\mathcal{F}|_U \rightarrow \mathcal{G}|_U$, so $s|_{V_i}$ refers to the induced natural transformation of functors $\mathcal{F}|_{V_i} \rightarrow \mathcal{G}|_{V_i}$ by only considering open subsets $W \subseteq V_i$. Thus, we have the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma(W, \mathcal{F}) & \longrightarrow & \prod \Gamma(W \cap V_i, \mathcal{F}) & \longrightarrow & \prod \Gamma(W \cap V_i \cap V_j, \mathcal{F}) \\ & & s(W) \downarrow & & \prod s(W \cap V_i) \downarrow & & \prod s(W \cap V_i \cap V_j) \downarrow \\ 0 & \longrightarrow & \Gamma(W, \mathcal{G}) & \longrightarrow & \prod \Gamma(W \cap V_i, \mathcal{G}) & \longrightarrow & \prod \Gamma(W \cap V_i \cap V_j, \mathcal{G}) \end{array}$$

hence $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ is a sheaf. $\square$

- 17. Skyscraper Sheaves.** Let X be a topological space, let P be a point, and let A be an abelian group. Define a sheaf $i_P(A)$ on X as follows: $i_P(A)(U) = A$ if $P \in U$, 0 otherwise. Verify that the stalk of $i_P(A)$ is A at every point $Q \in \overline{\{P\}}$, and 0 elsewhere. Hence, the name "Skyscraper sheaf." Show that this sheaf could also be described as $i_*(A)$, where A denotes the constant sheaf A on the closed subspace $\overline{\{P\}}$, and $i : \overline{\{P\}} \rightarrow X$ is the inclusion.

Proof. We first identify the restriction maps of $i_P(A)$. Let $U \subseteq V$ be open subsets of X , then $i_P(A)(V) \rightarrow i_P(A)(U)$ is the zero map if either V or U (hence V as well) does not contain P , and is the identity map if both V and U contain P . If $Q \in \overline{\{P\}}$, then every open neighborhood of Q contains P , so the direct system of open neighborhoods of Q induces the direct system consisting of a copy of A for every $U \ni Q$ with the identity map as transition maps, hence $(i_P(A))_Q = A$. If $Q \notin \overline{\{P\}}$, then there exists an open neighborhood W of Q not containing P , which means $i_P(A)(W) = 0$, hence $(i_P(A))_Q = 0$. The last statement is obvious. $\square$

- 18. Adjoint Property of f^{-1} .** Let $f : X \rightarrow Y$ be a continuous map of topological spaces. Show that for any sheaf $\mathcal{F}$ on X there is a natural map $f^{-1}f_*\mathcal{F} \rightarrow \mathcal{F}$, and for any sheaf $\mathcal{G}$ on Y there is a natural map $\mathcal{G} \rightarrow f_*f^{-1}\mathcal{G}$. Use these maps to show that there is a natural bijection of sets, for any sheaves $\mathcal{F}$ on X and $\mathcal{G}$ on Y ,

$$\text{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) = \text{Hom}_Y(\mathcal{G}, f_*\mathcal{F}).$$

Proof. Let U, V be open in X, Y , respectively. We unpack the definitions:

$$\begin{aligned}(f^{-1}f_*\mathcal{F})(U) &= \varinjlim_{W \supseteq f(U)} (f_*\mathcal{F})(W) = \varinjlim_{W \supseteq f(U)} \mathcal{F}(f^{-1}(W)), \\ (f_*f^{-1}\mathcal{G})(V) &= (f^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{W \supseteq f(f^{-1}(V))} \mathcal{G}(W).\end{aligned}$$

We also have

$$\begin{aligned}(f_*f^{-1}f_*\mathcal{F})(V) &= (f^{-1}f_*\mathcal{F})(f^{-1}(V)) \\ &= \varinjlim_{W \supseteq f(f^{-1}(V))} \mathcal{F}(f^{-1}(W)) \\ &= \mathcal{F}(f^{-1}(V)) \\ &= (f_*\mathcal{F})(V) \\ (f^{-1}f_*f^{-1}\mathcal{G})(U) &= \varinjlim_{W \supseteq f(U)} (f_*f^{-1}\mathcal{G})(W) \\ &= \varinjlim_{W \supseteq f(U)} \varinjlim_{W' \supseteq f(f^{-1}(W))} \mathcal{G}(W') \\ &= \varinjlim_{W \supseteq f(U)} \mathcal{G}(W) \\ &= (f^{-1}\mathcal{G})(U)\end{aligned}$$

Note that $V' := f(f^{-1}(V)) = V \cap f(X)$. An element of $(f^{-1}f_*\mathcal{F})(U)$ is of the form (W, s) where $W \supseteq f(U)$ and $s \in \mathcal{F}(f^{-1}(W))$. Define $\alpha : f^{-1}f_*\mathcal{F} \rightarrow \mathcal{F}$ as

$$\begin{aligned}\alpha(U) : (f^{-1}f_*\mathcal{F})(U) &\rightarrow \mathcal{F}(U) \\ (W, s) &\mapsto s|_U.\end{aligned}$$

An element of $(f_*f^{-1}\mathcal{G})(V)$ is of the form (W, t) where $W \supseteq V'$ and $t \in \mathcal{G}(W)$. Define $\beta : \mathcal{G} \rightarrow f_*f^{-1}\mathcal{G}$ as

$$\begin{aligned}\beta(V) : \mathcal{G}(V) &\rightarrow (f_*f^{-1}\mathcal{G})(V) \\ t &\mapsto (V, t).\end{aligned}$$

Let $\varphi : \mathcal{F} \rightarrow \mathcal{F}'$, $\psi : \mathcal{G} \rightarrow \mathcal{G}'$, where $\mathcal{F}, \mathcal{F}' \in \mathfrak{Sh}_X$, $\mathcal{G}, \mathcal{G}' \in \mathfrak{Sh}_Y$, $\varphi \in \text{Hom}_X(\mathcal{F}, \mathcal{F}')$, and $\psi \in \text{Hom}_Y(\mathcal{G}, \mathcal{G}')$. We explicitly describe the induced lifts with respect to f , namely $f_*\varphi : f_*\mathcal{F} \rightarrow f_*\mathcal{F}'$ and $f^{-1}\psi : f^{-1}\mathcal{G} \rightarrow f^{-1}\mathcal{G}'$. Let U, V be open sets in X, Y , respectively. Define

$$\begin{aligned}f_*\varphi(V) : f_*\mathcal{F}(V) &\rightarrow f_*\mathcal{F}'(V) \\ s &\mapsto \varphi(f^{-1}(V))(s)\end{aligned}$$

where $s \in \Gamma(f^{-1}(V), \mathcal{F})$, and define

$$\begin{aligned}f^{-1}\psi(U) : f^{-1}\mathcal{G}(U) &\rightarrow f^{-1}\mathcal{G}'(U) \\ (W, t) &\mapsto (W, \psi(W)(t)),\end{aligned}$$

where $(W, t) \in \varinjlim_{W \supseteq f(U)} \mathcal{G}(W)$. For every $\varphi : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$, we have the morphism $f_*\varphi : f_*f^{-1}\mathcal{G} \rightarrow f_*\mathcal{F}$, so we define $\beta^* : \text{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) \rightarrow \text{Hom}_Y(\mathcal{G}, f_*\mathcal{F})$ as

$$\beta^*\varphi = (f_*\varphi) \circ \beta.$$

Similarly, for every $\psi : \mathcal{G} \rightarrow f_*\mathcal{F}$, we have the morphism $f^{-1}\psi : f^{-1}\mathcal{G} \rightarrow f^{-1}f_*\mathcal{F}$, so we define $\alpha_* : \text{Hom}_Y(\mathcal{G}, f_*\mathcal{F}) \rightarrow \text{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F})$ as

$$\alpha_*\psi = \alpha \circ (f^{-1}\psi).$$

To show α_*, β^* are bijections with inverses to each other, it suffices to show $\alpha_* \circ \beta^* = \text{id}_{\text{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F})}$ and $\beta^* \circ \alpha_* = \text{id}_{\text{Hom}_Y(\mathcal{G} \rightarrow f_*\mathcal{F})}$:

$$(\alpha_* \circ \beta^*)(\varphi) = \alpha_*((f_*\varphi) \circ \beta) = \alpha \circ f^{-1}((f_*\varphi) \circ \beta) = \alpha \circ (f^{-1}f_*\varphi) \circ (f^{-1}\beta)$$

$$(\beta^* \circ \alpha^*)(\psi) = \beta^*(\alpha \circ (f^{-1}\psi)) = f_*(\alpha \circ (f^{-1}\psi)) \circ \beta = (f_*\alpha) \circ (f_*f^{-1}\psi) \circ \beta.$$

Again, we unpack the definitions. Let $(W, s) \in f^{-1}\mathcal{G}(\mathcal{U})$, where $f(\mathcal{U}) \subseteq W \subseteq Y$ with $t \in \Gamma(W, \mathcal{G})$. We want to show

$$(\alpha_*\beta^*\varphi)(\mathcal{U})((W, t)) = \varphi(\mathcal{U})((W, t)).$$

We have

$$\begin{aligned} (\alpha_*\beta^*\varphi)(\mathcal{U})((W, t)) &= (\alpha(\mathcal{U}) \circ (f^{-1}f_*\varphi))(\mathcal{U}) \circ (f^{-1}\beta)(\mathcal{U})((W, t)) \\ &= (\alpha(\mathcal{U}) \circ (f^{-1}f_*\varphi)(\mathcal{U}))((W, t)) \\ &= \alpha(\mathcal{U})((f^{-1}(W), \varphi(f^{-1}(W)(t)))) \\ &= \varphi(f^{-1}(W)((W, t)))|_{\mathcal{U}} \\ &= \varphi(\mathcal{U})((W, t)). \end{aligned}$$

Here, we used the fact that the restriction map commutes with $\varphi(\mathcal{U})$ by definition of morphisms between sheaves. Let $t \in \Gamma(V, \mathcal{G})$. We similarly have

$$\begin{aligned} (\beta_*\alpha^*\psi)(V)(t) &= ((f_*\alpha)(V) \circ (f_*f^{-1}\psi)(V) \circ \beta(V))(t) \\ &= ((f_*\alpha)(V) \circ (f_*f^{-1}\psi(V)))((V, t)) \\ &= (f_*\alpha)(V)(\psi(V)(t)) \\ &= \psi(V)(t). \end{aligned}$$

Hence, α_* and β^* are bijections with inverses to each other. □

22. Glueing Sheaves. Let X be a topological space, let $\mathcal{U} = \{\mathcal{U}_i\}$ be an open cover of X , and suppose we are given for each i a sheaf $\mathcal{F}_i$ on $\mathcal{U}_i$, and for each i, j an isomorphism $\varphi_{ij} : \mathcal{F}_i|_{\mathcal{U}_i \cap \mathcal{U}_j} \xrightarrow{\sim} \mathcal{F}_j|_{\mathcal{U}_i \cap \mathcal{U}_j}$ such that

- (1) for each i , $\varphi_{ii} = \text{id}$,
- (2) and for each i, j, k , $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $\mathcal{U}_i \cap \mathcal{U}_j \cap \mathcal{U}_k$.

Then there exists a unique sheaf $\mathcal{F}$ on X together with isomorphisms $\psi_i : \mathcal{F}|_{\mathcal{U}_i} \xrightarrow{\sim} \mathcal{F}_i$ such that for each i, j , $\psi_j = \varphi_{ij} \circ \psi_i$ on $\mathcal{U}_i \cap \mathcal{U}_j$. We say loosely that $\mathcal{F}$ is obtained by *glueing* the sheaves $\mathcal{F}_i$ via the isomorphisms φ_{ij} .

Shame on James for skipping this “standard exercise.”